SalvageShip

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1 Game Design Document

1.1 Overview

SalvageShip will be a single-player 3D sci-fi ship salvaging game. The player is an intern for a massive interstellar corporation with near-limitless resources called Serenity Ltd flying in a small company spacecraft on the way to the corporate headquarters in a distant star system. After an asteroid damages the ship's engine, the player crash lands on a remote planet near a larger crashed company cargo vessel.

The company's onboard support terminal successfully locates a lightspeed drive component inside the wreck to repair the player's ship. Being an intern, however, the player does not have access to emergency tools required to get to the part of the crashed ship that contains the component. With some basic tools and a limited backpack, the player dismantles accessible cargo, furniture and ship equipment and transfers the recovered materials to a recycler present on the player's ship and receives credits in return.

The player can spend these credits on the company's paid replicator, also present on the player's ship, to obtain better tools, increase inventory capacity, deploy Asset Transfer Crates, and gain access to deeper sections of the cargo vessel with these upgrades. Upon reaching the vessel's core and recovering the component, the player can repair their own ship and escape the planet.

The game will be similar to ***Hardspace: Shipbreaker*** (Blackbird Interactive, 2022, Figure 1) and ***Ship Graveyard Simulator 2*** (Games Incubator, 2023, Figure 2)



Figure 1. Hardspace: Shipbreaker from steam page



Figure 2. Ship Graveyard Simulator 2 from steam page

1.2 Game Mechanics and Features

1.2.1 Interactions, Rules, and Resources

1.2.1.1 Player Movement and Salvaging

The player moves through the ship and limited environment outside the ship using standard WASD controls, with mouse movement for camera. They can walk through connected rooms and corridors and interact with salvageable objects. When the player aims at an object that can be dismantled with an unlocked tool, the object has a visible outline and a prompt to interact.

If it is an object that can be dismantled but with a different tool, the object will have a red outline, and the prompt will tell the player which tool is needed if they continue looking at the object for a certain amount of time.

Holding the interact button begins the dismantling process. A progress bar will show how much time remains, and the player must remain close and continue holding the interact button until the progress bar is finished. Walking too far or releasing the button cancels the interaction. Once dismantled, the object is removed from the environment and gets added to the player's backpack.

The player starts with a basic multi tool after completing the tutorial that can pick up small loose items and dismantle some low value objects. More advanced tools will allow the player to dismantle more complicated objects which are worth more.

1.2.1.2 Backpack and Inventory

The player has a backpack built into their spacesuit with a limited carrying capacity. Every object has a weight value and credit value. Opening the inventory will display the current capacity, maximum capacity, and estimated value of the stored salvage.

The player cannot dismantle an object if doing so would exceed the maximum capacity. When the pack is full, the player must transfer the contents to the recycler on their ship or to an Asset Transfer Crate unlockable throughout the cargo ship.

The player will be able to upgrade the backpack to have increased storage and a more accurate price estimate. The player will also be able to buy 'visor' upgrades that will let them see the estimated value of an item before picking it up to strategically choose which items to disassemble and see the current storage and maximum storage on the normal HUD.

1.2.1.3 Recycler and Replicator

The player's ship contains two primary stations, a recycler and a replicator. The recycler accepts the salvage stored in the player's backpack and converts it into company credits.

The replicator functions as the game's upgrade store. It can create equipment in exchange for credits, including tool upgrades, backpack expansions, visor upgrades, and Asset Transfer Crates.

1.2.1.4 Salvage Tools and Upgrades

Tool upgrades determine which objects and barriers the player can dismantle. Each newly unlocked tool can also dismantle objects handled by lower level tools so the player does not need to switch tools manually.

The basic multi tool can recover loose cargo and open crates, dismantle furniture and other unsecured objects. A power drill allows the player to remove bolted, bulkier furniture, lockers, lights, and simple electronics. An angle grinder allows the player to dismantle things like reinforced doors, pipes, and other metal objects. A plasma cutter allows the player to dismantle large machinery and eventually extract the component needed to repair their ship.

All tools can also be upgraded to decrease disassemble time. Other upgrades include backpack capacity, 'visor' HUD upgrades, and a flashlight for dark areas. All upgrades are purchased with credits

1.2.1.5 Asset Transfer Crates

In each new section of the cargo ship, the player can purchase an Asset Transfer Crate from the replicator and deploy it in specific marked areas near the entrance of the section. Each area supports one crate.

The player can deposit salvage from their backpack into any crate, allowing them to continue to explore new sections without having to go all the way back out to the player's ship. The only replicator will be on the player's ship, however.

1.2.1.6 Ship Areas and Progression

The game starts in a small cargo area on the player's ship as a tutorial, where the game instructs the player to sell some of the small things they can just pick up on their own ship to afford the basic multi tool and open a stuck door, allowing them to continue to the cargo ship.

The first section of the cargo ship is the cargo hold, containing lots of loose cargo and basic furniture the player can collect immediately, as well as some higher-level cargo the player cannot salvage at the beginning.

The next area is the crew quarters, containing bedrooms, bathrooms, dining areas, and similar rooms. Much of this furniture is bolted and requires a power drill.

The next area is the maintenance corridor, unlocked with an angle grinder, which allows the

player to disassemble reinforced panels, pipes, and sealed doors, along with some previously inaccessible things in earlier areas.

The final section is the engineering bay, containing generators, heavy machinery, and large computer systems, requiring the plasma cutter to disassemble. At the farthest reaches of this area lies the lightspeed drive core, where the player needs to purchase a special one-time-use tool to extract the component needed for their ship.

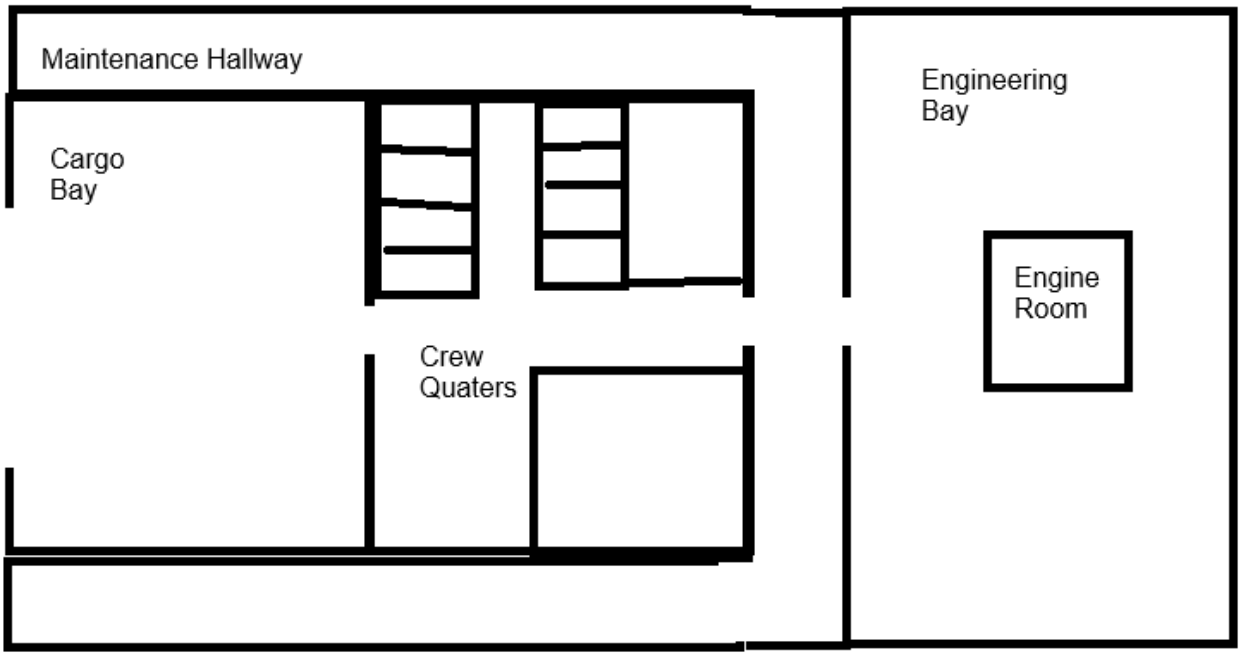


Figure 3. Ship Floorplan Diagram

1.2.1.7 Credits and Salvage Value

Credits are the primary currency. Every salvageable object has a value based on size, rarity, and tool requirement. Optional high-value objects will be placed throughout the ship to reward careful exploration, but the main path ensures enough objects to earn enough credits to finish the game without having to fully dismantle everything.

1.2.2 Objectives

The player's main objective is to recover a component for their lightspeed drive from the core of a crashed cargo vessel. After recovering the component, the player installs it in their ship and escapes the planet.

The player also has optional high-value objects throughout the ship. The player can view found objects alongside silhouettes of undiscovered ones in a menu on the pause menu.

The game can be completed without finding these, but it is an extra challenge.

There are also completion percentages for each area and the ship as a whole. These also are not necessary to get through the game but again is a challenge for the player should they wish to undertake it.

1.2.3 Conflict

The main conflict comes from Serenity Ltd.'s unwillingness to share their resources with a low level intern without payment. The player is limited by their current tools and upgrades. The further areas are also more difficult to explore.

1.2.4 Menus and Interfaces

1.2.4.1 HUD

The first-person HUD provides information needed for gameplay. It displays the player's current credits, tools, and backpack capacity once that upgrade is unlocked. It also shows the completion percentage for the ship area the player is currently exploring.

When the player aims at a salvageable object, an interaction prompt shows the object and its estimated value when that upgrade has been purchased. Holding the interaction key displays a progress bar while the object is being dismantled. Objects that require a higher-level tool will say which tool is required and will have a different outline color.

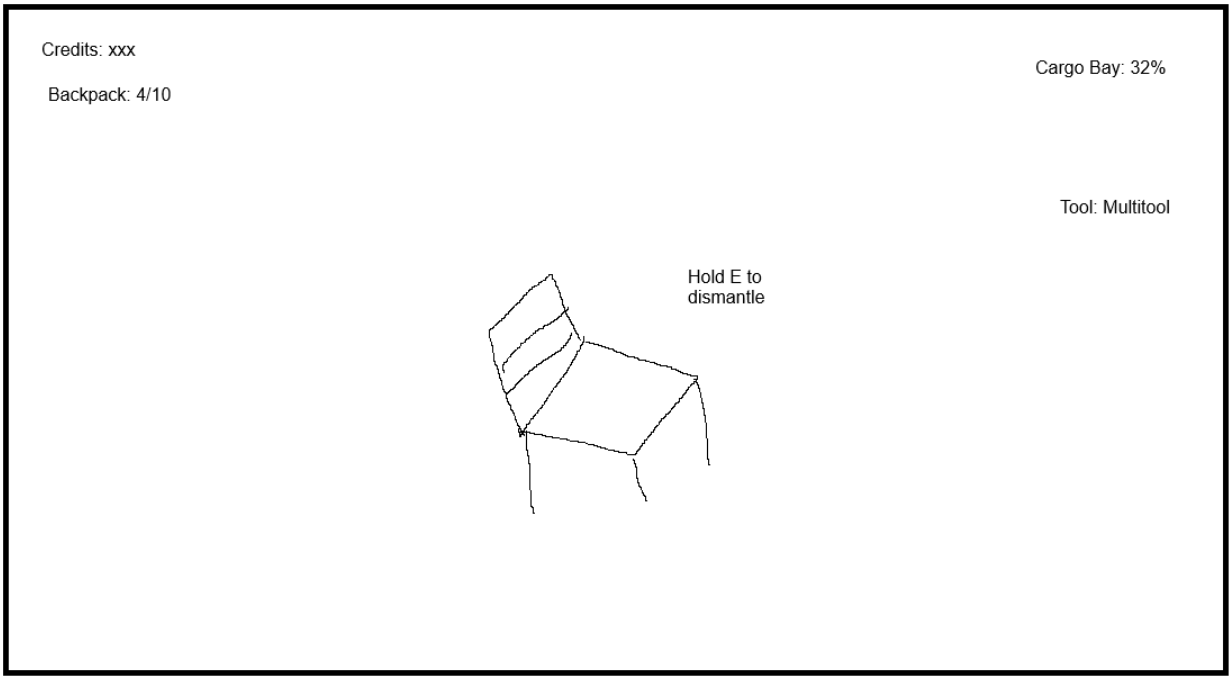


Figure 4. HUD

1.2.4.2 Recycler

The recycler screen allows the player to transfer their collected salvage to the recycler and receive credits in return based on the value of the objects

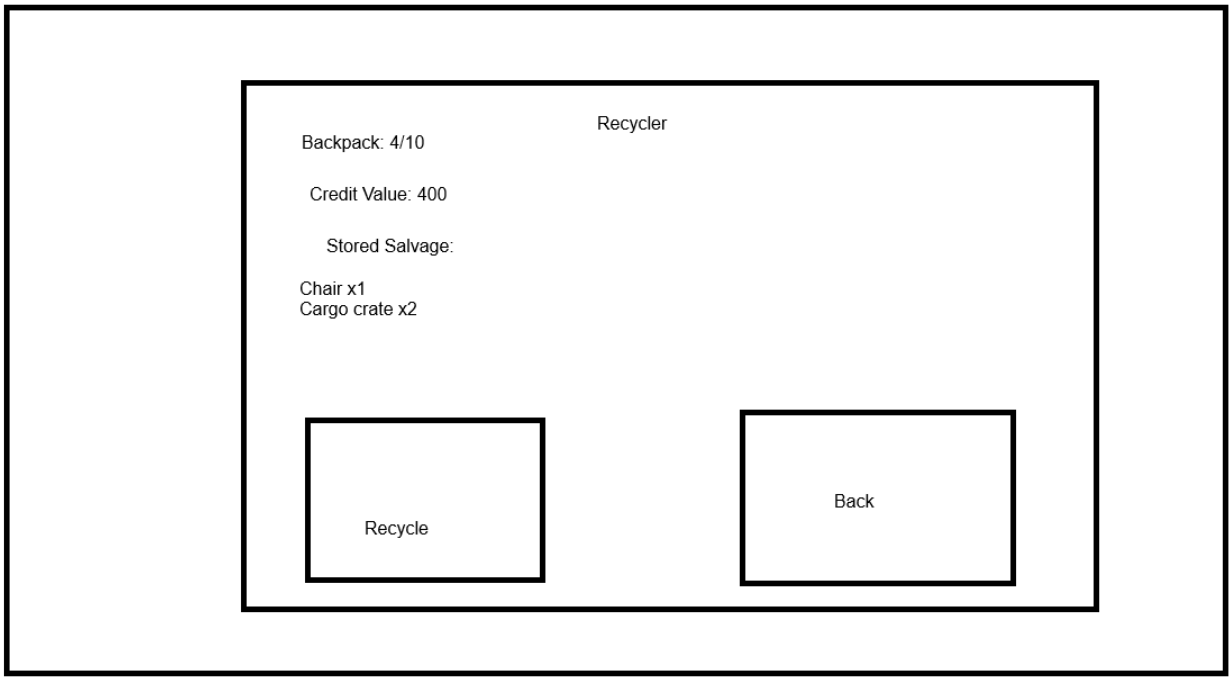


Figure 5. Recycler UI

1.2.4.3 Replicator Menu

The replicator menu lists available tools and their upgrades, pack upgrades, visor upgrades, flashlight, and other equipment. Locked upgrades display their cost.

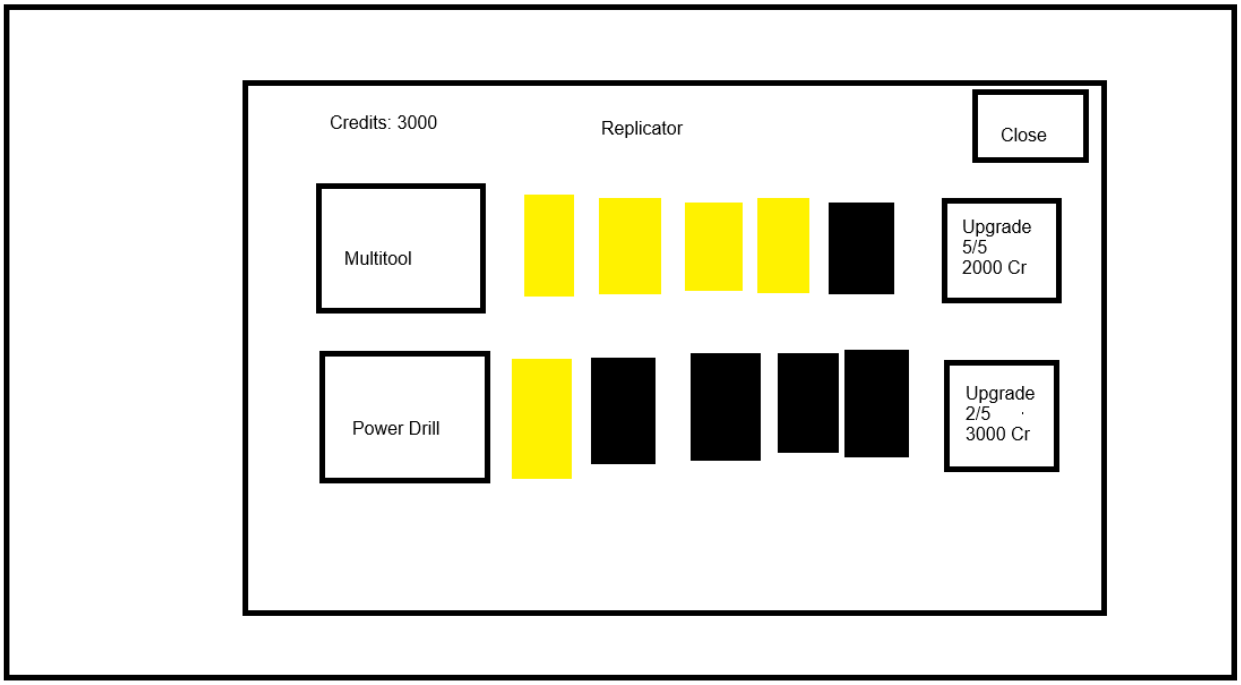


Figure 6. Replicator UI

1.2.4.4 Pause

The pause menu allows the player to open settings, resume, or quit the game. It also lets them view overall completion percentage across the whole vessel. There is also a popup that displays the special high value objects.

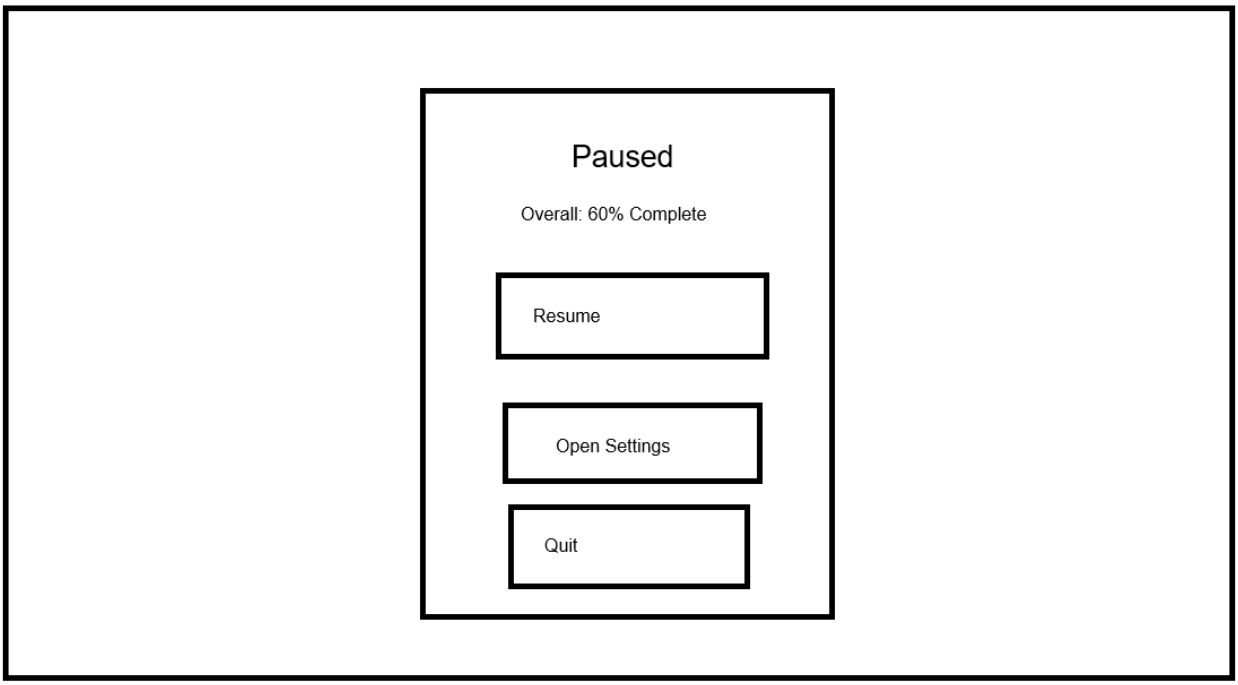


Figure 7. Pause Menu

1.2.4.5 Settings Menu

The settings menu includes options of music volume, sfx volume, display settings, control sensitivity, and control mappings / remapping.

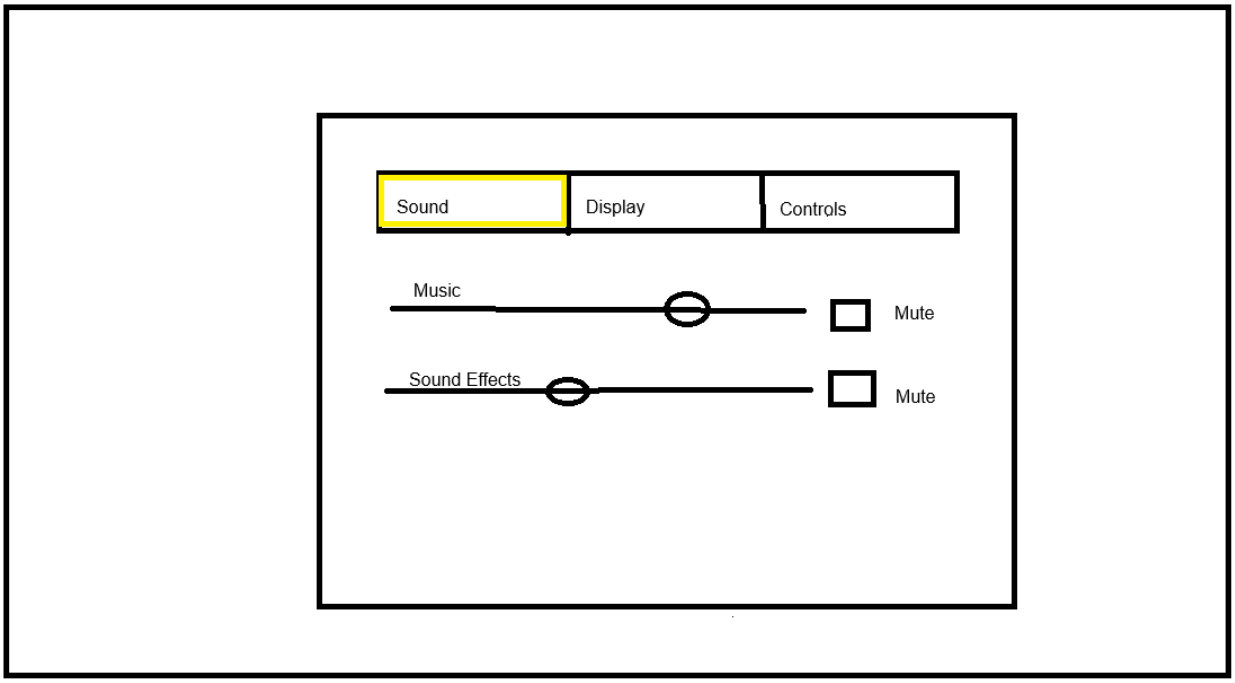


Figure 8. Settings Menu

1.2.4.6 Tutorial Prompts

Tutorial prompts appear throughout the game, but more so near the beginning that explain game mechanics to the player as they unlock them.

1.2.4.7 End Screen

After the player repairs their ship, an end screen displays the player's final completion percentage and allows them to quit to the main menu or continue exploring to increase the percentage.



Figure 9. End Screen

1.2.5 Characters and story

The player controls an intern employed by Serenity Ltd, an intergalactic corporation that values profit above everything else – including employee safety. After their small spacecraft crash lands on a remote planet, the player learns that a nearby cargo vessel contains a rare lightspeed drive component needed to repair their spaceship.

Serenity Ltd.'s automated support terminal communicates with the player through tutorial prompts and corporate messages. The company knows how to save the player but requires the player to pay for every tool and upgrade needed to succeed.

2 Technical Design Document and Art List

2.1 Game Engine

SalvageShip will be developed using Godot engine 4. Scripts will be written in GDScript

2.2 Other Software and Tools

GitHub will be used for version control and backups.

Blender will be used to create and animate original 3D models.

GIMP will be used to edit textures, UI images, and other 2D assets.

2.3 Platform

The game will be released as a Windows executable.

2.4 Scene State Diagram

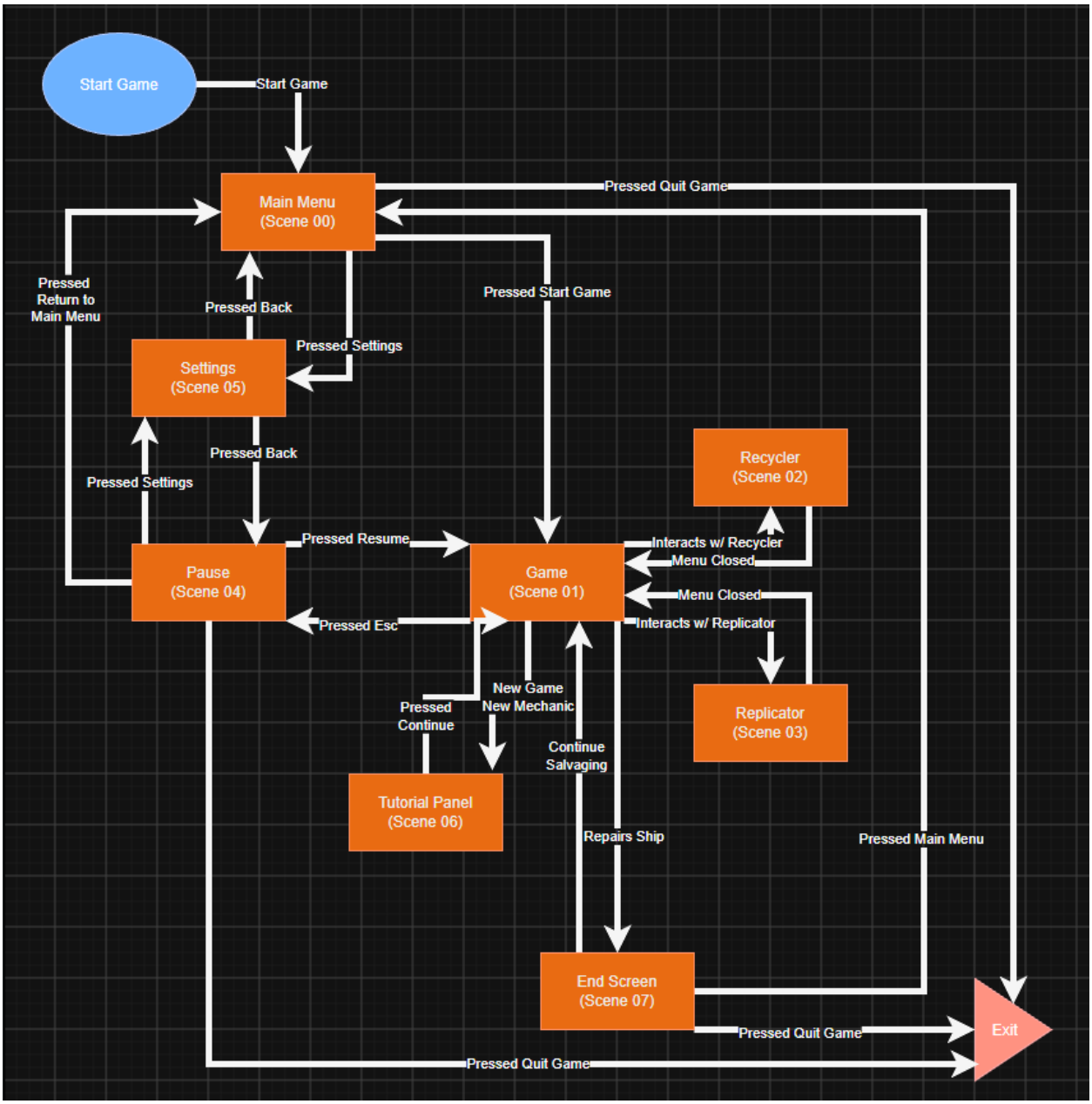


Figure 10. Scene State Diagram

2.5 Scene List

2.5.1 Main Menu

Scene 00

Objectives	Provides the player with a way to start the game, open the settings, or quit the game.
Outgoing Transitions	TO Game WHEN player presses Start Game TO Settings Menu WHEN player presses Settings TO Exit WHEN player presses Quit Game
Game Objects	Main Menu UI Title Text Background Art Start Game Button Settings Button Quit Game Button Menu Music
Comments	n/a

2.5.2 Game

Scene 01	
Objectives	Contains the main bulk of the gameplay. The player can explore the crashed cargo vessel, dismantle objects, earn credits, and purchase upgrades
Outgoing Transitions	TO Recycler Menu WHEN player interacts with recycler TO Replicator Menu WHEN player interacts with replicator TO Pause Menu WHEN player presses escape TO Tutorial Panel WHEN a new mechanic is unlocked TO End Screen WHEN player repairs their ship
Game Objects	Game Manager Player Player Camera Player Ship Cargo Vessel Recycler Replicator Asset Transfer Crates HUD Music Employee Support Terminal Locked Barriers Lightspeed Drive Component
Comments	There will also be many small game objects including furniture, loose items, computer panels, and machinery.

2.5.3 Recycler Menu

Scene 02	
Objectives	Allows the player to transfer salvage from their backpack to the recycler and receive credits
Outgoing Transitions	TO Game WHEN player closes menu
Game Objects	Recycler UI Stored Salvage List Total Value Label Transfer Salvage Button Close Button
Comments	Opened from the recycler in the player ship.

2.5.4 Replicator Menu

Scene 03	
Objectives	Allows the player to purchase new salvage tools, speed upgrades, backpack capacity upgrades, visor upgrades, flashlights, and Asset Transfer Crates
Outgoing Transitions	TO Game WHEN player closes menu
Game Objects	Replicator UI Credits Label Tool Purchase Buttons Upgrade Purchase Buttons Locked Upgrade Labels Close Button
Comments	Opened from the replicator inside the player ship.

2.5.5 Pause Menu

Scene 04	
Objectives	Pauses gameplay and can resume, view ship completion progress, open settings, return to the main menu, or quit the game.
Outgoing Transitions	TO Game WHEN player presses resume TO Settings Menu WHEN player presses settings TO Main Menu WHEN player presses Return to Main Menu TO Exit WHEN player presses Quit Game
Game Objects	Pause Menu UI Overall Completion Label Resume Button

	Settings Button Return to Main Menu Button Quit Game Button
Comments	Can be opened at any point by pressing esc

2.5.6 Settings Menu

Scene 05	
Objectives	Allows the player to adjust music volume, sound effects volume, display settings, mouse sensitivity, and control mappings
Outgoing Transitions	TO Main Menu WHEN player closes after opening settings from Main Menu TO Pause Menu WHEN player closes after opening settings from Pause
Game Objects	Settings UI Music Volume Slider Sound Effect Volume Slider Display Settings Controls Mouse Sensitivity Slider Control Mapping Buttons Back Button
Comments	

2.5.7 Tutorial Panel

Scene 06	
Objectives	Introduces control and mechanics to the player at the beginning of a new game and when new features are unlocked throughout gameplay
Outgoing Transitions	TO Game WHEN player closes tutorial prompt
Game Objects	Tutorial UI Instruction Text Continue Button
Comments	

2.5.8 End Screen

Scene 07	
Objectives	Congratulates the player after repairing their ship and displays the final ship completion percentage

Outgoing Transitions	TO Game WHEN player presses Continue Salvaging TO Main Menu WHEN player presses Main Menu TO Exit WHEN player presses Quit Game
Game Objects	End Screen UI Completion Label Continue Salvaging Button Main Menu Button Quit Game Button
Comments	

2.6 Game Object / Component Diagrams

2.6.1 Main Menu

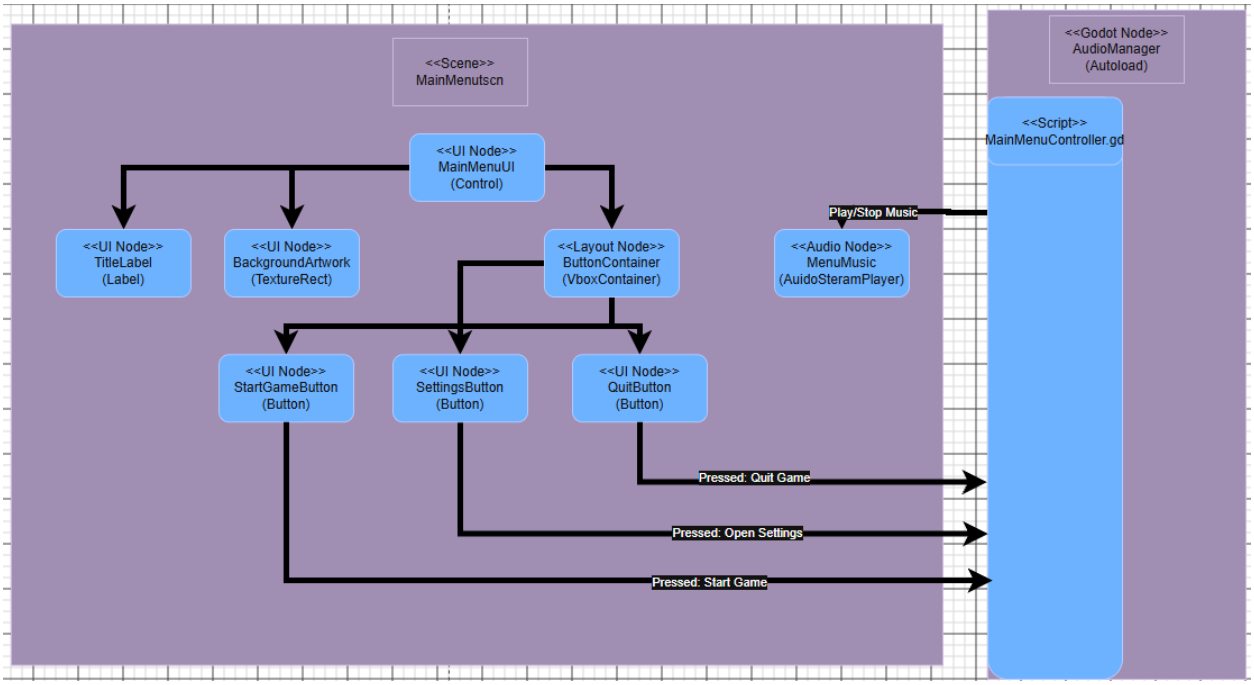


Figure 11. Main Menu Game Object / Component Diagram

2.6.2 Game

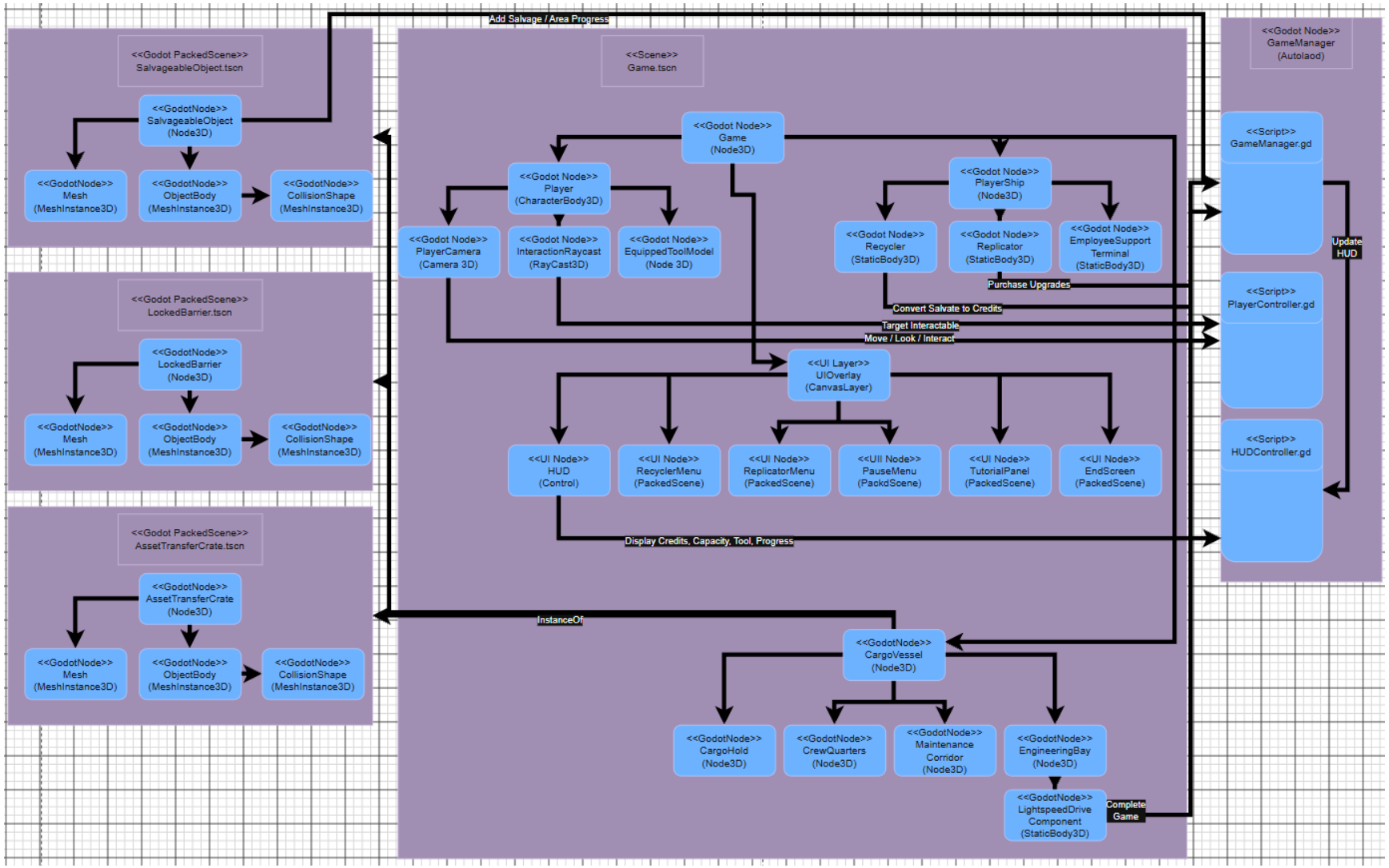


Figure 12. Game Scene Game Object / Component Diagram

2.6.3 Recycler Menu

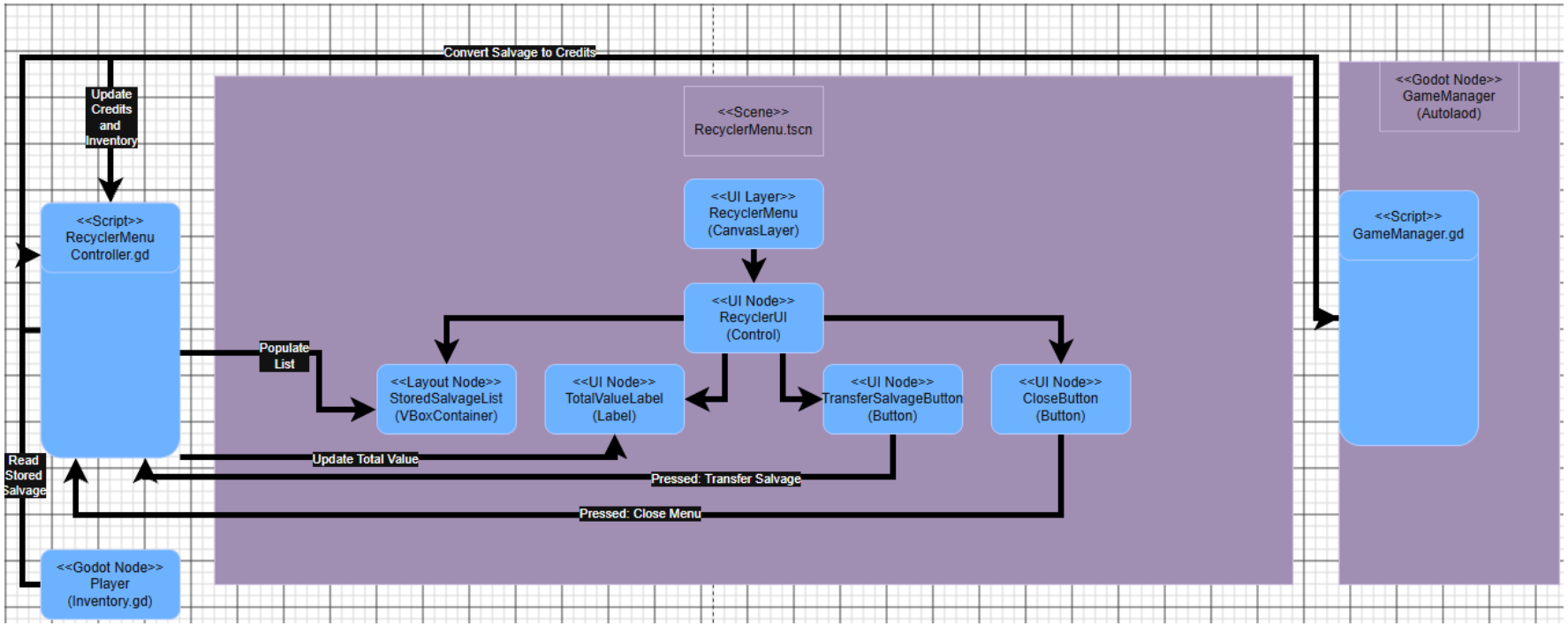


Figure 13. Recycler Menu Game Object / Component Diagram

2.6.4 Replicator Menu

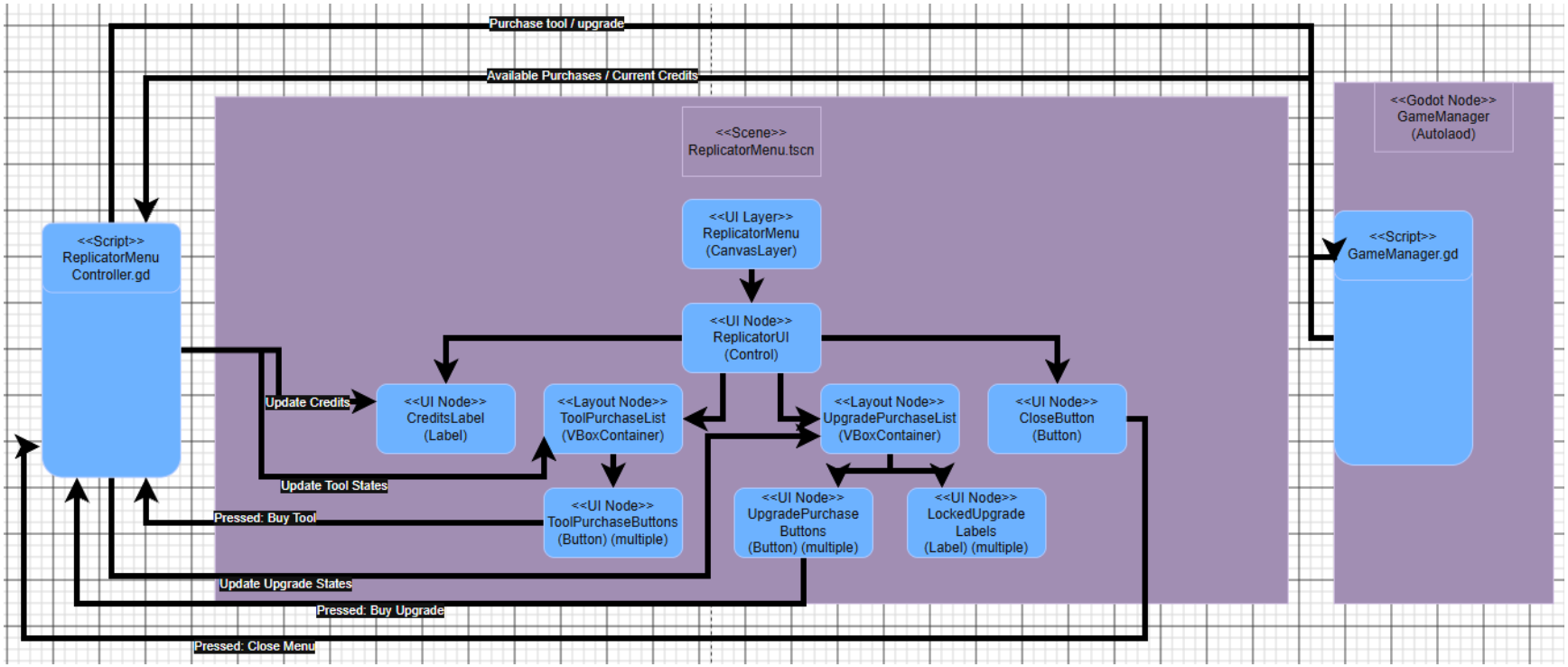


Figure 14. Replicator Menu Game Object / Component Diagram

2.6.5 Pause Menu

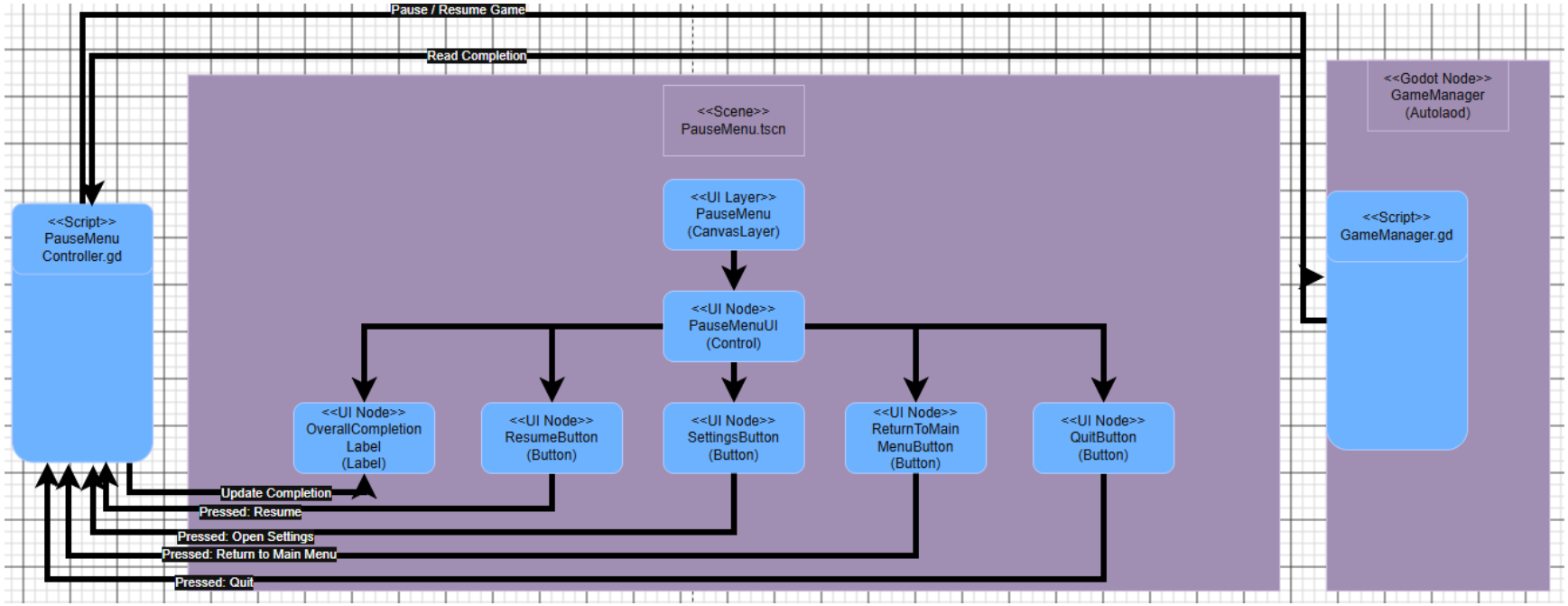


Figure 15. Pause Menu Game Object / Component Diagram

2.6.6 Settings Menu

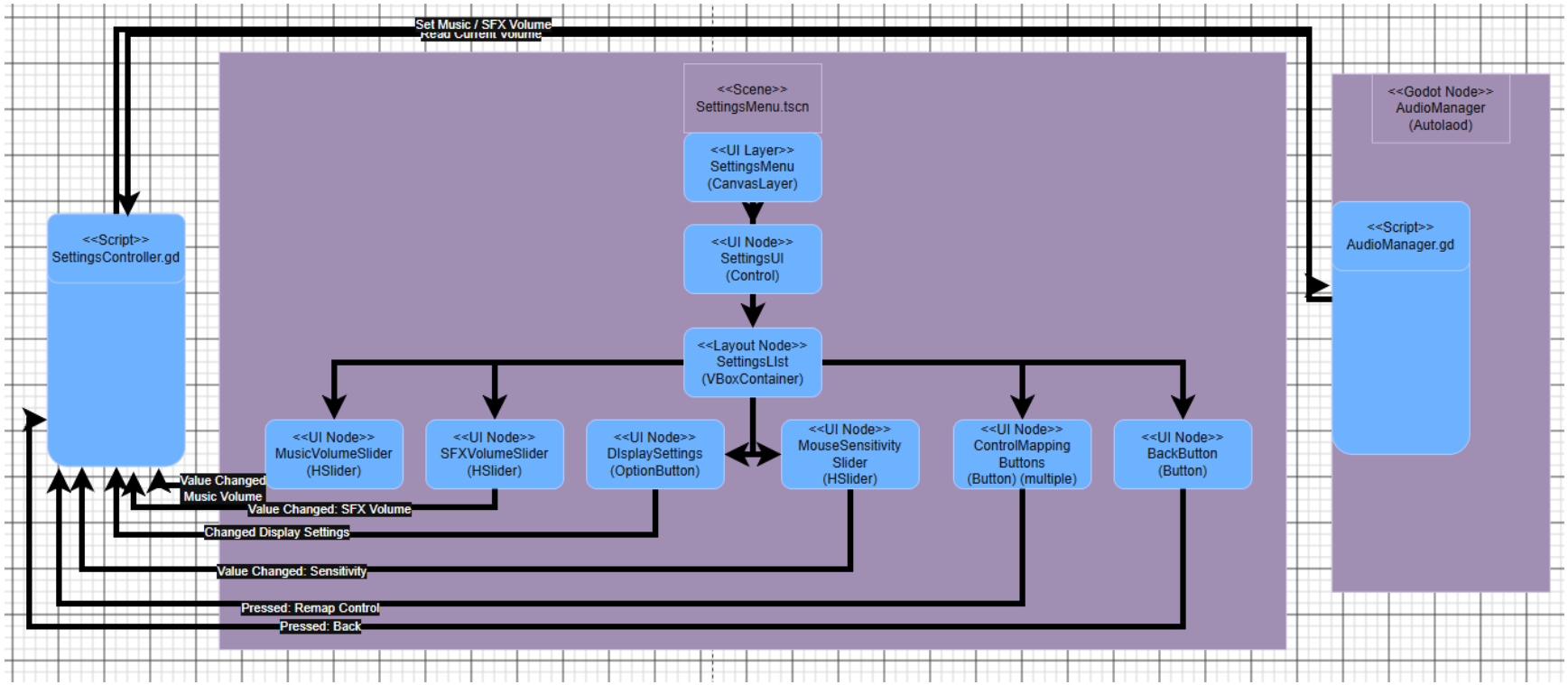


Figure 16. Settings Menu Game Object / Component Diagram

2.6.7 Tutorial Panel

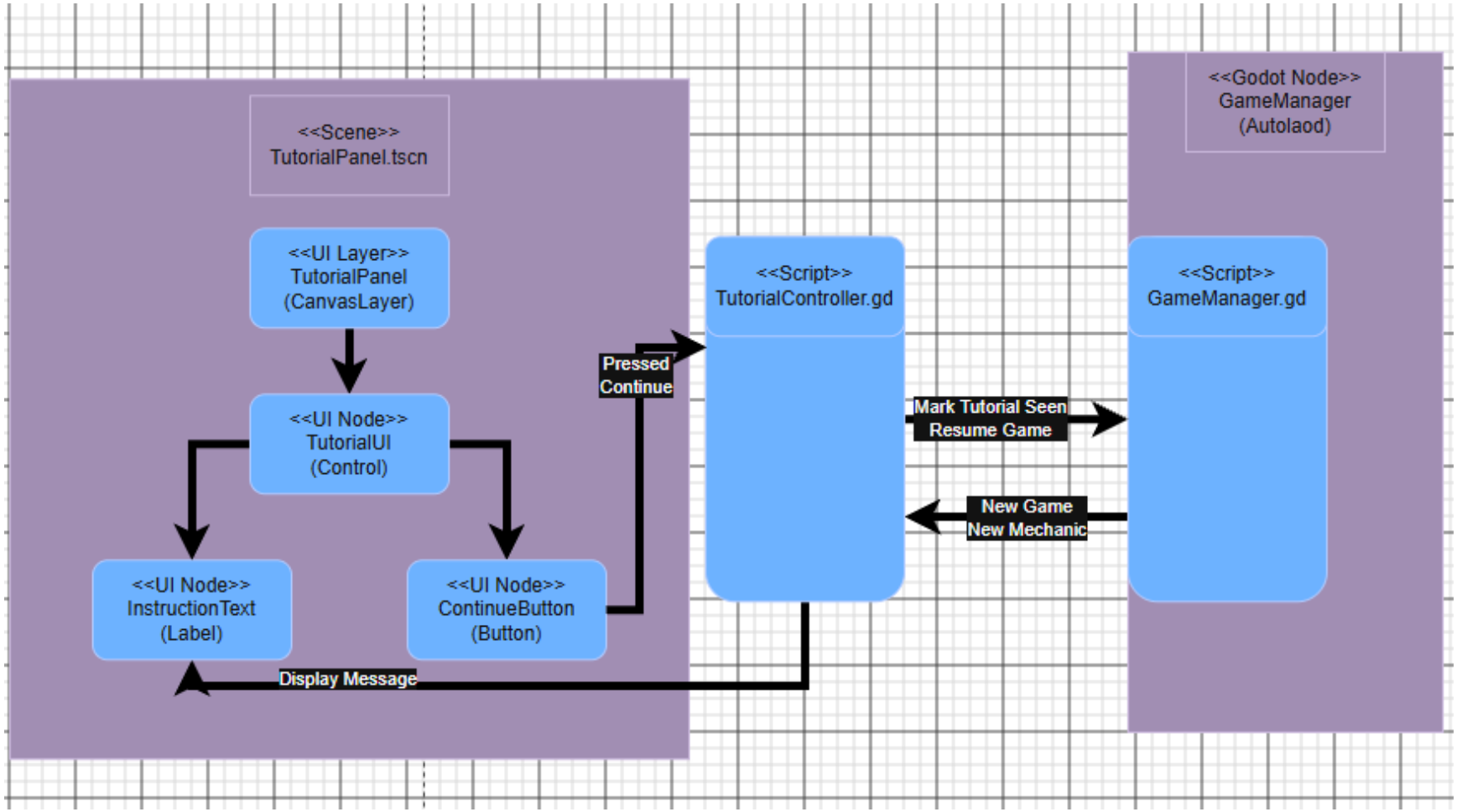


Figure 17. Tutorial Panel Game Object / Component Diagram

2.6.8 End Screen

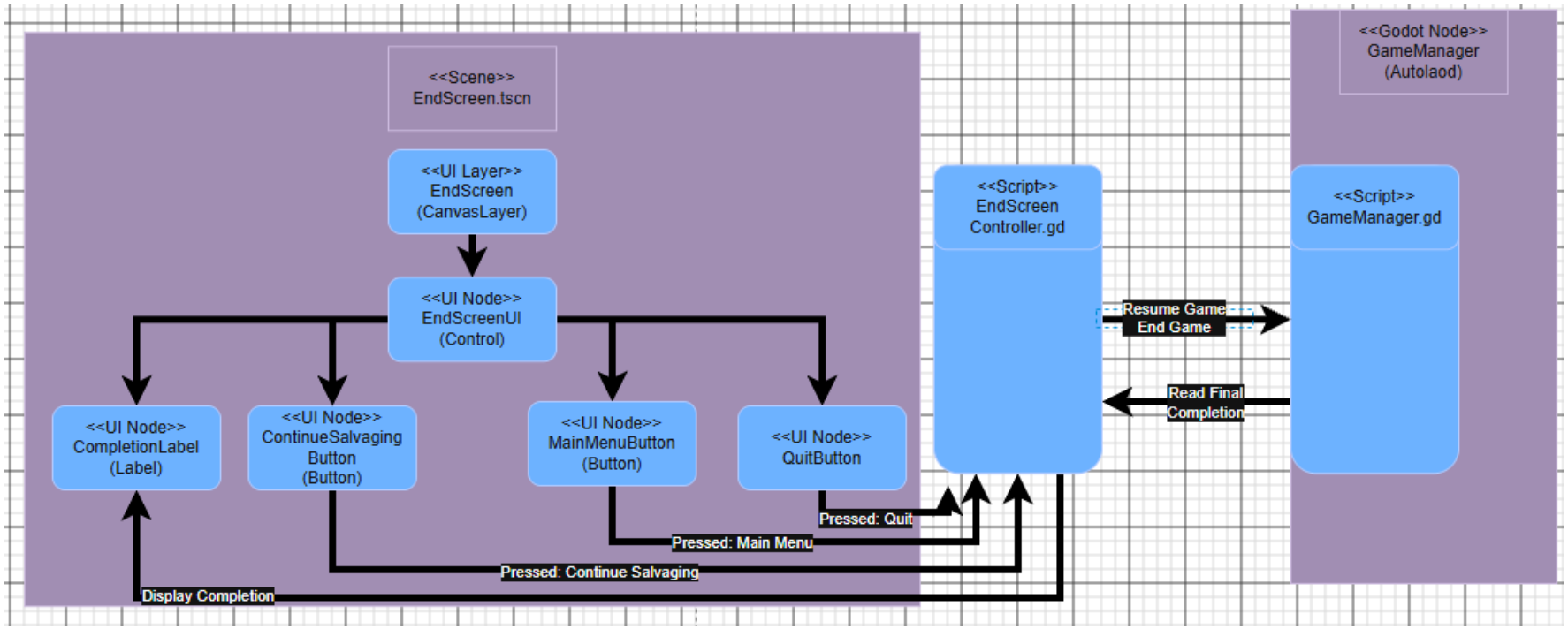


Figure 18. End Screen Game Object / Component Diagram

2.7 Game Object List

2.7.1 Game Manager

Game Manager (Autoload)	
Description	Stores and manages persistent gameplay data, such as credits, inventory, owned upgrades and tools, and completion percentages
Tags	Game Data Persistent
Required Components	Node GameManager.gd
Related Game Objects	
Parent	n/a
Children	n/a
Associations	Player Salvageable Object Recycler Replicator HUD Pause Menu UI End Screen UI
Comments	Keeps player progress between scenes

2.7.2 Player

Player	
Description	The first-person character controlled by the player. They can move through the vessel, aim at objects, dismantle salvage, and interact with machines.
Tags	Player Controllable
Required Components	CharacterBody3D CollisionShape3D PlayerController.gd Inventory.gd
Related Game Objects	
Parent	Game
Children	Player Camera Interaction Raycast Equipped Tool Model
Associations	Salvageable Object Locked Barrier

	Recycler Replicator Asset Transfer Crate HUD
Comments	Use keyboard and mouse

2.7.3 Player Camera

Player Camera	
Description	Provides first person view and detects objects targeted by the player
Tags	Camera Interaction
Required Components	Camera3D RayCast3D InteractionRaycast.gd
Related Game Objects	
Parent	Player
Children	n/a
Associations	Salvageable Object Locked barrier Recycler Replicator HUD
Comments	InteractionRaycast identifies objects that can be dismantled or interacted with

2.7.4 Player Ship

Player Ship	
Description	The player's spaceship and the starting area. Contains the recycler, replicator, employee support terminal, and the engine to repair
Tags	Environment Starting Area
Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D
Related Game Objects	
Parent	Game
Children	Recycler Replicator Employee Support Terminal
Associations	Player LightSpeed Drive component

	Recycler Menu Replicator Menu
Comments	Functions as the player's home base

2.7.5 Cargo Vessel

Cargo Vessel	
Description	The crashed Serenity Ltd. Cargo vessel containing salvageable objects and the component needed to repair the player's ship
Tags	Environment Main Level
Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D
Related Game Objects	
Parent	Game
Children	Cargo Hold Crew Quarters Maintenance Corridor Engineering Bay
Associations	Player Salvageable Object Locked Barrier Asset Transfer Crate
Comments	Contains all primary gameplay areas for exploration

2.7.6 Cargo Hold

Cargo Hold	
Description	The first accessible section of the cargo vessel, containing loose cargo, unsecured furniture and other basic items
Tags	Environment Area 1
Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D
Related Game Objects	
Parent	Cargo Vessel
Children	Salvageable Objects Locked Barriers Asset Transfer Crate Area
Associations	Player Basic Multi tool

	HUD Area Progress Data
Comments	

2.7.7 Crew Quarters

Crew Quarters	
Description	The second section of the cargo vessel. Contains living spaces, lockers, bolted furniture, and more valuable salvage
Tags	Environment Area 2
Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D
Related Game Objects	
Parent	Cargo Vessel
Children	Salvageable Objects Locked Barriers Asset Transfer Crate Area
Associations	Player Power Drill HUD Area Progress Data
Comments	

2.7.8 Maintenance Corridor

Maintenance Corridor	
Description	The third section of the cargo vessel. It contains reinforced panels, pipes, sealed doors, and routes to previously inaccessible areas
Tags	Environment Area 3
Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D
Related Game Objects	
Parent	Cargo Vessel
Children	Salvageable Objects Locked Barriers Asset Transfer Crate Area
Associations	Player Angle Grinder HUD

	Area Progress Data
Comments	

2.7.9 Engineering Bay

Engineering Bay	
Description	The final section of the cargo vessel. It contains heavy machinery, generators, large computer systems, the lightspeed drive component.
Tags	Environment Area 4 Final Area
Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D
Related Game Objects	
Parent	Cargo Vessel
Children	Salvageable Objects Locked Barriers Lightspeed Drive Component Asset Transfer Crate Area
Associations	Player Plasma Cutter Lightspeed Drive Component End Screen
Comments	

2.7.10 Salvageable Object

Salvageable Object (PackedScene)	
Description	A reusable object type the player can dismantle to receive salvage value. There are many things that use this, such as loose cargo, furniture, lockers, electronics, pipes, panels, and machinery.
Tags	Salvageable PackedScene Interactable
Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D Salvageable.gd
Related Game Objects	
Parent	Player Ship Cargo Hold

	Crew Quarters Maintenance Corridor Engineering Bay
Children	Outline Effect Tool Requirement Icon
Associations	Player Player Camera Game Manager HUD Recycler
Comments	

2.7.11 Locked Barrier

Locked Barrier	
Description	A door or other obstruction that prevents players from accessing later areas until the tool required for that area is unlocked.
Tags	Barrier Progression Interactable
Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D LockedBarrier.gd
Related Game Objects	
Parent	Cargo Vessel
Children	Lock Icon Interaction Prompt
Associations	Player Player Camera Replicator Salvage Tools
Comments	

2.7.12 Recycler

Recycler	
Description	A machine in the player ship that converts stored salvage in the player's backpack to credits
Tags	Machine Interactable
Required Components	Node3D StaticBody3D CollisionShape3D

	MeshInstance3D Recycler.gd
Related Game Objects	
Parent	Player Ship
Children	Interaction Prompt
Associations	Player Game Manager Recycler Menu
Comments	

2.7.13 Replicator

Replicator	
Description	A machine inside the player ship, allowing the player to spend credits on tools, upgrades, flashlights, and Asset Transfer Crates
Tags	Machine Interactable Upgrade Store
Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D Replicator.gd
Related Game Objects	
Parent	Player Ship
Children	Interaction Prompt
Associations	Player Game Manager Replicator Menu Asset Transfer Crate
Comments	

2.7.14 Asset Transfer Crate

Asset Transfer Crate (PackedScene)	
Description	A deployable container purchased from the replicator that can be placed at certain marked locations throughout the ship. The player can use it to deposit salvage instead of walking all the way back to the ship
Tags	PackedScene Interactable Storage
Required Components	Node3D StaticBody3D

	CollisionShape3D MeshInstance3D AssetTransferCrate.gd
Related Game Objects	
Parent	Cargo Hold Crew Quarters Maintenance Corridor Engineering Bay
Children	Interaction Prompt
Associations	Player Game Manager Recycler Replicator
Comments	

2.7.15 Lightspeed Drive Component

Lightspeed Drive Component	
Description	A rare component needed to repair the player's ship and complete the game
Tags	Quest Item Objective
Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D LightspeedDrive.gd
Related Game Objects	
Parent	Engineering Bay
Children	Interaction Prompt
Associations	Player Plasma Cutter Player Ship End Screen
Comments	

2.7.16 Employee Support Terminal

Employee Support Terminal	
Description	An automated Serenity Ltd. terminal that provides messages, story information, and tutorial instructions to the player
Tags	Story Tutorial Interactable

Required Components	Node3D StaticBody3D CollisionShape3D MeshInstance3D SupportTerminal.gd
Related Game Objects	
Parent	Player Ship
Children	Interaction Prompt Tutorial Screen
Associations	Player Tutorial UI Game Manager
Comments	

2.7.17 HUD

HUD	
Description	Displays important gameplay information, such as credits, backpack capacity, equipped tool, area completion percentage, interaction prompts, and dismantle progress
Tags	UI Persistent
Required Components	CanvasLayer Control HUDController.gd
Related Game Objects	
Parent	Game
Children	Credits Label Backpack Capacity Label Equipped Tool Label Area Completion Label Interaction Prompt Dismantle Progress Bar
Associations	Player Player Camera Game Manager Salvageable Object
Comments	

2.7.18 Main Menu UI

Main Menu UI	
Description	Displays the game title and allows the player to start the game, open settings, or quit

Tags	UI Menu
Required Components	CanvasLayer Control MainMenuController.gd
Related Game Objects	
Parent	Main Menu
Children	Background Artwork Title Text New Game Button Settings Button Quit Game Button
Associations	Settings UI Game Audio Manager
Comments	

2.7.19 Pause Menu UI

Pause Menu UI	
Description	Pauses gameplay and allows the player to resume, view completion progress, open settings, return to menu, or quit
Tags	UI Menu Pause
Required Components	CanvasLayer Control PauseMenuController.gd
Related Game Objects	
Parent	Game
Children	Overall Completion Label Resume Button Settings Button Return to Main Menu Button Quit Game Button
Associations	Game Manager Settings UI Main Menu UI
Comments	

2.7.20 Settings UI

Settings UI	
Description	Allows the player to change music volume, sound effects volume, display options, mouse sensitivity, and control keybindings

Tags	UI Menu Settings
Required Components	CanvasLayer Control SettingsController.gd
Related Game Objects	
Parent	Main Menu Pause Menu
Children	Music Volume Slider Sound Effect Volume Slider Display Settings Controls Mouse Sensitivity Slider Control Mapping Buttons Back Buttons
Associations	Main Menu UI Pause Menu UI Audio Manager
Comments	Returns to the menu it was opened from

2.7.21 Recycler UI

Recycler UI	
Description	Displays the salvage currently carried by the player and allows the player to transfer to the recycler for credits
Tags	UI Menu Recycler
Required Components	CanvasLayer Control RecyclerMenuController.gd
Related Game Objects	
Parent	Game
Children	Salvage List Total Value Label Transfer Salvage Button Close Button
Associations	Recycler Player Game Manager HUD
Comments	

2.7.22 Replicator UI

Replicator UI	
Description	Displays the available tools and upgrades the player can purchase with credits
Tags	UI Menu Upgrade Store
Required Components	CanvasLayer Control ReplicatorMenuController.gd
Related Game Objects	
Parent	Game
Children	Credits Label Tool Purchase Buttons Upgrade Purchase Buttons Locked Upgrade Labels Close Button
Associations	Replicator Player Game Manager Asset Transfer Crate
Comments	

2.7.23 Tutorial UI

Tutorial UI	
Description	Displays instructional prompts when the player begins a new game or encounters a new mechanic
Tags	UI Tutorial
Required Components	CanvasLayer Control TutorialController.gd
Related Game Objects	
Parent	Game
Children	Instruction Text Continue Button
Associations	Corporate Support Terminal Player Game Manager
Comments	

2.7.24 End Screen UI

End Screen UI	
Description	Displays the completion message after the player repairs their ship and shows the final completion percentage
Tags	UI Menu End Game
Required Components	CanvasLayer Control EndScreenController.gd
Related Game Objects	
Parent	Game
Children	Completion Label Continue Salvaging Button Main Menu Button Quit Game Button
Associations	Lightspeed Drive Player Ship Game Manager Main Menu UI
Comments	

2.7.25 Audio Manager

Audio Manager (Autoload)	
Description	Controls background music, menu music, gameplay sound effects, and volume settings
Tags	Audio Persistent
Required Components	Node AudioStreamPlayer AudioManager.gd
Related Game Objects	
Parent	n/a
Children	Music Player Sound Effect Player
Associations	Settings UI Main Menu UI Game HUD
Comments	

2.8 Component List

2.8.1 GameManager.gd

GameManager.gd (Custom)	
Description	Stores persistent game data, such as credits, inventory, owned tools, upgrades, and area completion percentages
Requires	Node Configured as Autoload
Comments	

2.8.2 PlayerController.gd

PlayerController.gd (Custom)	
Description	Controls first-person player movement, mouse look, and player interaction
Requires	CharacterBody3D Player Camera
Comments	Attached to player object

2.8.3 Inventory.gd

Inventory.gd (Custom)	
Description	Tracks the salvage objects stored in the player's backpack, total value, and the capacity
Requires	Player Game Manager
Comments	Prevents the player from collecting additional salvage when the pack is full

2.8.4 InteractionRaycast.gd

InteractionRaycast.gd (Custom)	
Description	Detects object targeted by player's camera and identifies if it can be dismantled or interacted with
Requires	Camera3D RayCast3D HUD
Comments	Updates the interaction prompt and outlines the currently targeted object

2.8.5 Salvageable.gd

Salvageable.gd (Custom)

Description	Stores a salvage object's value, required tool tier, dismantle duration, and area completion contribution
Requires	Salvageable Object Game Manager Player
Comments	Removes the object from the world and adds it to the player's inventory

2.8.6 LockedBarrier.gd

LockedBarrier.gd (Custom)	
Description	Controls barriers such as doors and sealed panels that require a specific tool to remove
Requires	Locked Barrier Player Game Manager
Comments	Displays the required tool when player targets the barrier

2.8.7 Recycler.gd

Recycler.gd (Custom)	
Description	Opens the Recycler UI and converts the salvage stored in the player's backpack into credits
Requires	Recycler Recycler UI Game Manager
Comments	

2.8.8 Replicator.gd

Replicator.gd (Custom)	
Description	Opens the Replicator UI and allows the player to purchase tools, upgrades and Asset Transfer Crates
Requires	Replicator Replicator UI Game Manager
Comments	

2.8.9 AssetTransferCrate.gd

AssetTransferCrate.gd (Custom)	
Description	Allows the player to transfer stored salvage from their pack through an active Asset Transfer Crate
Requires	Asset Transfer Crate Player Game Manager

Comments	Provides a closer location to deposit salvage in each area instead of having to go all the way back to the ship.

2.8.10 LightspeedDrive.gd

LightspeedDrive.gd (Custom)	
Description	Controls the final Lightspeed Drive Component and triggers the end sequence after the player repairs their ship.
Requires	Lightspeed Drive Component Player Ship Game Manager
Comments	

2.8.11 SupportTerminal.gd

SupportTerminal.gd (Custom)	
Description	Displays messages, tutorial information, and story text
Requires	Employee Support Terminal Tutorial UI
Comments	

2.8.12 HUDController.gd

HUDController.gd (Custom)	
Description	Updates the HUD's credits, backpack capacity, equipped tool, current area completion percentage, interaction prompt, and dismantling progress bar
Requires	HUD Player Game Manager
Comments	

2.8.13 MainMenuController.gd

MainMenuController.gd (Custom)	
Description	Handles Main Menu button input and starts the game, opens settings, or quits the application
Requires	Main Menu UI Settings UI
Comments	

2.8.14 PauseMenuController.gd

PauseMenuController.gd (Custom)	
Description	Pauses and resumes gameplay, displays overall completion progress, and handles pause menu button input

Requires	Pause Menu UI Game Manager Settings UI
Comments	Opens when escape pressed during gameplay

2.8.15 SettingsController.gd

SettingsController.gd (Custom)	
Description	Controls audio volume, display options, mouse sensitivity, and other settings
Requires	Settings UI Audio Manager Godot audio buses
Comments	Returns the player to the menu from which settings was opened

2.8.16 RecyclerMenuController.gd

RecyclerMenuController.gd (Custom)	
Description	Displays the player's stored salvage, calculates its total value, and transfers it to the recycler to exchange for credits
Requires	Recycler UI Player Game Manager
Comments	

2.8.17 ReplicatorMenuController.gd

ReplicatorMenuController.gd (Custom)	
Description	Displays available purchases, checks credit requirements, and applies tool and upgrade purchases
Requires	Replicator UI Game Manager
Comments	

2.8.18 TutorialController.gd

TutorialController.gd (Custom)	
Description	Displays tutorial prompts on the start of a new game or when the player unlocks a new mechanic
Requires	Tutorial UI Game Manager
Comments	

2.8.19 EndScreenController.gd

EndScreenController.gd (Custom)

Description	Displays the final completion percentage and handles the options to continue playing, go to main menu, and quit the game
Requires	End Screen UI Game Manager
Comments	

2.8.20 AudioManager.gd

AudioManager.gd (Custom)	
Description	Controls background music, menu music, gameplay sound effects, and volume settings
Requires	AudioStreamPlayer Audio buses
Comments	Autoload

2.8.21 Node

Node (Provided by Godot)	
Description	A basic node used for organization
Requires	
Comments	

2.8.22 Node3D

Node3D (Provided by Godot)	
Description	The 3D version of a node with position, rotation, and scale
Requires	
Comments	Root of all 3D game objects and environments

2.8.23 CharacterBody3D

CharacterBody3D (Provided by Godot)	
Description	A Godot 3D physics body for player controlled character movement
Requires	CollisionShape3D
Comments	

2.8.24 Camera3D

Camera3D (Provided by Godot)	
Description	Renders the 3D environment from a viewpoint
Requires	
Comments	

2.8.25 RayCast3D

RayCast3D (Provided by Godot)

Description	Casts a ray to the game world to detect objects
Requires	CollisionShape3D
Comments	

2.8.26 StaticBody3D

StaticBody3D (Provided by Godot)	
Description	A physics body for immovable environment objects
Requires	CollisionShape3D
Comments	

2.8.27 CollisionShape3D

CollisionShape3D (Provided by Godot)	
Description	Defines the physical collision shape for a 3D object
Requires	Shape
Comments	

2.8.28 MeshInstance3D

MeshInstance3D (Provided by Godot)	
Description	Displays a 3D mesh in the game world
Requires	Mesh
Comments	Used for ships, rooms, machines, props, and salvageable objects

2.8.29 CanvasLayer

CanvasLayer (Provided by Godot)	
Description	Draws child interface elements above the game world
Requires	
Comments	

2.8.30 Control

Control (Provided by Godot)	
Description	A base node for interface elements that support anchors, layout, and input
Requires	
Comments	

2.8.31 Label

Label (Provided by Godot)	
Description	Displays text on user interface elements
Requires	Font
Comments	

2.8.32 Button

Button (Provided by Godot)	
Description	An interface control that has a pressed state
Requires	
Comments	

2.8.33 ProgressBar

ProgressBar (Provided by Godot)	
Description	Displays a value as a progress bar
Requires	
Comments	

2.8.34 HSlider

HSlider (Provided by Godot)	
Description	A horizontal slider interface control
Requires	
Comments	

2.8.35 AudioStreamPlayer

AudioStreamPlayer (Provided by Godot)	
Description	Plays music and sound effect audio streams
Requires	Audio stream Audio bus
Comments	

2.8.36 PackedScene

PackedScene (Provided by Godot)	
Description	A scene that can be instantiated multiple times
Requires	
Comments	

2.8.37 AnimationPlayer

AnimationPlayer (Provided by Godot)	
Description	Plays animations for nodes
Requires	Animation
Comments	

2.9 Art List

2.9.1 Static 3D Assets

2.9.1.1 Player Ship

Player Ship	
Description	A small Serenity Ltd. Spacecraft that crash landed on a planet next to a cargo vessel.
Format	Blender .blend file
Comments	n/a
Concept Art	Capsule art

2.9.1.2 Cargo Vessel Exterior

Cargo Vessel Exterior	
Description	The exterior model of the Serenity Ltd. Cargo vessel located near the player ship, with damaged hull sections, and open cargo hold
Format	Blender .blend file
Comments	n/a
Concept Art	Capsule art

2.9.1.3 Cargo Vessel Interior

Cargo Vessel Interior	
Description	The interior portion of the cargo vessel, containing all the explorable areas the player can traverse.
Format	Blender .blend files
Comments	n/a
Concept Art	Figure 3

2.9.1.4 Salvageable Prop Set

Salvageable Prop Set	
Description	A reusable set of 3D props that can be dismantled for credits, including loose cargo, crates, chairs, tables, electronics, pipes, machines, computers, and beds
Format	Blender .blend files
Comments	There will be many variations
Concept Art	

2.9.1.5 Salvage Tool Set

Salvage Tool Set	
Description	Player-held 3D tools, a multi tool, power drill, angle grinder, and plasma cutter.
Format	Blender .blend files
Comments	
Concept Art	

2.9.1.6 Player Ship Terminals

Player Ship Terminals	
Description	Interactable terminals within the player ship. There will be models for the recycler, replicator, and employee support terminal
Format	Blender .blend files
Comments	
Concept Art	

2.9.1.7 Asset Transfer Crate

Asset Transfer Crate	
Description	A deployable container to transfer salvage from the player's backpack from inside the cargo ship
Format	Blender .blend file
Comments	
Concept Art	

2.9.1.8 Locked Barrier Set

Locked Barrier Set	
Description	Sealed doors, or otherwise blocked pathways that can only be opened once the player has purchased a specific tool
Format	Blender .blend file
Comments	
Concept Art	

2.9.1.9 Lightspeed Drive Component

Lightspeed Drive Component	
Description	A distinct component located deep inside the cargo ship. It is the final item needed to repair the player's ship.
Format	Blender .blend file
Comments	
Concept Art	

2.9.2 Animated 3D Assets

2.9.2.1 Replicator Animation

Replicator Animation	
Description	An animation that plays after the player purchases an upgrade
Format	Blender .blend file
Animated Sequence	1. Idle. The machine makes small movements while idle 2. Fabricate. The machine moves around and creates the item 3. Dispense. The machine moves the item to the player once complete
Comments	
Concept Art	

2.9.3 Static 2D Assets

2.9.3.1 Main Menu and Background

Main Menu	
Description	A title image for SalvageShip and a background showing the player ship with the crashed cargo vessel
Format	Background PNG RGBA, 1920 x 1080 px.
Comments	Displayed behind the main menu UI
Concept Art	

2.9.3.2 HUD

HUD	
Description	Interface icons for credits, backpack capacity, equipped tool, completion percentages, and other information
Format	PNG RGBA. Multiple 32 x 32 images
Comments	
Concept Art	Figure 4

2.9.3.3 Menu Panels and Buttons

Menu Panels and Buttons	
Description	UI panels, buttons, sliders, and overlays used by Main Menu, Pause Menu, Recycler Menu, Replicator Menu, Tutorial UI, and End Screen
Format	PNG RGBA, Scalable nine-slice panels and buttons
Comments	
Concept Art	Figures 5 - 9

2.10 Audio Asset List

2.10.1 Background Music

Background Music	
Description	A looping ambient sci-fi industrial music track that plays during gameplay
Format	OGG audio file, stereo
Comments	

2.10.2 Menu Button Click

Menu Button Click	
Description	A short UI click sound played when the player selects a button
Format	OGG audio file
Comments	

2.10.3 Salvage Dismantling

Salvage Dismantling	
Description	A mechanical sound played while the player holds the interaction key to dismantle a salvageable object
Format	OGG audio file
Comments	

2.10.4 Invalid Interaction

Invalid Interaction	
Description	A brief warning sound played when the player attempts to dismantle an object that requires a locked tool or when their backpack is full
Format	OGG audio file
Comments	

2.10.5 Recycler

Recycler	
Description	A recycling sound followed by a confirmation sound when the credits get transferred to the player.
Format	OGG audio file
Comments	

2.10.6 Upgrade Purchase

Upgrade Purchase	
Description	A fabrication sound plays while the replicator makes the upgrade, and a confirmation chime when it is completed
Format	OGG audio file
Comments	

2.10.7 Asset Transfer Crate

Asset Transfer Crate	
Description	A container opening and transfer sound played when the player deposits salvage into an Asset Transfer Crate
Format	OGG audio file
Comments	

2.10.8 Support Terminal Notification

Support Terminal Notification	
Description	A short electronic chime played when the Employee Support Terminal displays a new tutorial prompt or story message
Format	OGG audio file

Comments	

3.0 Task Lists

All tasks will be completed by Zachary Sansone

Week	Team Tasks	Programming Tasks	Art / Audio Tasks	Testing TODO
Week 00	• Finish and submit GDD • Create project schedule	• Create Godot project and GitHub repo • Test some Godot 3D nodes and first person camera things	• Create concept arts • Test in blender	• NA
Week 01	• Review scope and cut features if necessary	• Create main menu, settings, pause, tutorial, and start the game scene • Implement a first person controller, camera, raycast, and basic HUD • Add a few basic salvageable objects and add the interaction • Implement the player's backpack	• Create a prototype player ship model • Create a prototype cargo vessel exterior • Create a few low poly props • Create basic HUD and UI assets	• Test movement, camera, collisions, interaction, and dismantling • Test menu buttons and pause/settings
Week 02	• Get some feedback from people on core gameplay loop	• Implement salvage values, credits, recycler, and replicator • Add the tool requirements for objects • Add purchasable upgrades • Flesh out the cargo hold	• Create recycler, replicator and employee support terminal • Create the tools • Create the recycler and replicator menus	• Test the salvage collection, inventory, recycler, and upgrades • Make sure the core loop is functional
Week 03	• Review progression pacing	• Implement the tools • Add the locked barriers • Add Asset Transfer Crates • Add completion tracking • Protype of the further areas	• Create further ship areas • Create the barriers • Create the remaining tools	• Test all the tools and their interactions • Test Asset Transfer Crates • Test overall game progression through the areas

Week 04	• Get more feedback	• Implement the lightspeed drive component and ship repairing • Create the end screen • Implement the tutorial and Employee Support Terminal • Add Animations	• Create animations • Create the Lightspeed Drive component • Create remaining UI assets	• Test a full playthrough • Test tutorial triggers • Test the win condition • Test menus • Test completion percentages
Week 05		• Add save and loading • Fix bugs and balance • Replace placeholder content	• Add music and sound effects • Polish the environment	• Same test again
Week 06	• Submit	• Fix bugs • Finalize placement of assets	• Finish the rest of the assets	• Do a full playthrough with everything in place • Test the executable

4.0 Appendix

4.0.1 AI usage disclaimer

During the course of this assignment, my USC Chat GPT account was used to brainstorm ideas for certain concepts, generate the capsule image on the title page, and to proofread the document for any spelling mistakes.